

The Necessity of Exploitation in Bayesian Optimization: A Systematic Benchmark of Four Acquisition Strategies on Six Problems

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Abstract

Bayesian optimization (BO) is widely used for expensive black-box functions, where a Gaussian process (GP) surrogate drives an acquisition function that balances exploration of uncertain regions against exploitation of promising ones. A fundamental open question is whether the GP’s uncertainty estimate alone—without any exploitation signal—is sufficient to outperform blind random search.

I address this question with a systematic benchmark of four acquisition strategies: Expected Improvement (EI), Upper Confidence Bound (UCB), Uncertainty (maximum posterior standard deviation), and Random. All four strategies share the same normalized GP surrogate and are evaluated on six problems spanning dimensions 2 to 10, mixing classical synthetic functions with physics-based engineering test functions. Each strategy–problem pair is repeated over 10 independent seeds, yielding 240 BO runs total. A Friedman–Nemenyi non-parametric test ($N = 60$ blocks, $k = 4$) provides statistically robust comparisons.

The Friedman test strongly rejects equal performance ($\chi^2(3) = 128.82$, $p = 9.73 \times 10^{-28}$). Nemenyi post-hoc comparisons reveal two statistically separated equivalence classes: balanced strategies {EI, UCB} significantly outperform pure-exploration strategies {Uncertainty, Random} (all cross-class $p < 10^{-6}$), while Uncertainty and Random are indistinguishable from each other ($p = 0.974$). This Uncertainty–Random tie demonstrates that exploitation of the GP’s posterior mean—not the uncertainty estimate alone—is the necessary condition for sample-efficient BO. A secondary finding is that GP input normalization to known problem bounds is a first-class engineering requirement on multi-scale problems: omitting it degrades BO to random-search performance on Piston (7D), while leaving the Random baseline unaffected, isolating the fault to the surrogate.

1 Introduction

Bayesian optimization (BO) is the standard approach to optimizing expensive black-box functions: objectives that

are costly to evaluate, return no gradients, and may be non-convex or multimodal (Frazier, 2018; Shahriari et al., 2016). Applications range from hyperparameter tuning in machine learning to aerodynamic shape optimization, where each function evaluation can take minutes to hours of computation or physical experiment. BO sequentially builds a probabilistic surrogate model—typically a Gaussian process (GP)—and uses an *acquisition function* to decide where to evaluate next. The acquisition function must balance *exploitation* (querying where the surrogate predicts high values) against *exploration* (querying where the surrogate is uncertain, to reduce the risk of missing a better region).

A natural question is whether the GP’s uncertainty estimate alone is sufficient to drive sample efficiency. Purely uncertainty-driven sampling—selecting the point of maximum posterior standard deviation—uses the GP’s calibration to target information gain, without any regard for whether the point is predicted to have a high objective value. If uncertainty-driven sampling outperforms blind random search, it would suggest that the GP’s uncertainty model, rather than the exploitation signal, is the primary engine of BO’s advantage. If, on the other hand, it performs no better than random, it would implicate exploitation as the necessary component. Classical acquisition functions such as Expected Improvement (EI) and Upper Confidence Bound (UCB) combine both signals; understanding which component matters more is fundamental to designing and interpreting BO algorithms.

I address this question through a systematic benchmark. I evaluate four strategies—EI, UCB, Uncertainty (maximum posterior standard deviation), and Random—on six benchmark problems spanning dimensions 2 to 10 and two problem families: classical synthetic functions and physics-based engineering test functions. All strategies share the same GP surrogate and BO loop, so any performance difference is attributable solely to the acquisition function. I use a Friedman–Nemenyi non-parametric statistical framework (Demšar, 2006) to draw conclusions robust to non-Gaussian regret distributions.

Contributions.

1. **Systematic 4-strategy benchmark.** I run 4 strategies $\times$ 6 problems $\times$ 10 seeds = 240 BO runs under identical conditions and analyze results with a non-parametric test (Friedman + Nemenyi) rather than mean-only comparisons. The analysis reveals two statistically separated equivalence classes: balanced strategies {EI, UCB} and pure-exploration strategies {Uncertainty, Random}.
2. **GP normalization is necessary on multi-scale engineering problems.** An experimental comparison on Piston (7D, inputs spanning approximately seven orders of magnitude) shows that omitting GP input normalization degrades BO to random-search performance, while the Random baseline is unaffected. This isolates the fault to the surrogate and validates the choice of explicit, bounds-based normalization.
3. **Exploitation, not uncertainty, is the critical ingredient.** The statistical equivalence of Uncertainty and Random (Nemenyi $p = 0.974$) provides a controlled decomposition: the GP’s posterior variance, used without the posterior mean, provides no measurable advantage over blind Sobol sampling. Exploitation of the GP mean is the necessary condition for sample-efficient BO.

The remainder of the paper is organized as follows. Section 2 reviews the BO framework, the GP surrogate, and related acquisition function families. Section 3 details the experimental design, surrogate implementation, and statistical methodology. Section 4 presents all benchmark results. Section 5 interprets the findings and discusses limitations. Section 6 concludes.

2 Background

2.1 The Bayesian Optimization Loop

Bayesian optimization solves $\max_{x \in \mathcal{X}} f(x)$ when f is a black-box function that is expensive to evaluate, returns no gradients, and may be multimodal (Frazier, 2018). The algorithm proceeds in two alternating steps. First, a *surrogate model* is fitted to all observations $\{(x_i, y_i)\}_{i=1}^n$; it provides a posterior distribution over f with a calibrated mean $\mu(x)$ and variance $\sigma^2(x)$. Second, an *acquisition function* $\alpha(x)$ converts the posterior into a scalar score at each candidate point; the next evaluation location is $x_{n+1} = \arg \max_{x \in \mathcal{X}} \alpha(x)$. Solving the inner maximization is cheap because α is an analytical function of the posterior.

BO is best suited to problems where: (i) each evaluation of f is expensive (minutes to hours), (ii) the dimensionality is low-to-moderate, and (iii) gradients of f are unavailable (Frazier, 2018; Shahriari et al., 2016).

2.2 Gaussian Process Surrogate

A Gaussian process (GP) is a distribution over functions fully specified by a mean function $m(x)$ and a positive-definite kernel $k(x, x')$ (Frazier, 2018). After observing n points, the GP posterior at a new location x_* is Gaussian with

$$\mu(x_*) = k_*^\top (K + \sigma_n^2 I)^{-1} \mathbf{y}, \quad \sigma^2(x_*) = k(x_*, x_*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_*,$$

where $K_{ij} = k(x_i, x_j)$, $k_* = [k(x_i, x_*)]_i$, and σ_n^2 is the noise variance. The key computational cost is the $n \times n$ matrix inversion (or Cholesky decomposition), which is $\mathcal{O}(n^3)$ and limits exact GP inference to $n \lesssim 10^3$ observations (Shahriari et al., 2016).

The kernel encodes smoothness assumptions about f . I use the radial basis function (RBF) kernel with *automatic relevance determination* (ARD): one independent length-scale ℓ_j per input dimension j . ARD allows the GP to learn which dimensions carry most of the variation in f , effectively performing soft variable selection. Kernel hyperparameters are fitted by maximizing the exact marginal log-likelihood at each BO iteration.

2.3 Acquisition Functions

Acquisition functions trade off exploitation of high-mean regions against exploration of high-variance regions. All classical acquisition functions use both $\mu(x)$ and $\sigma(x)$; the key difference is how they weight the two signals.

Expected Improvement (EI). $\alpha_{\text{EI}}(x) = \mathbb{E}[\max(f(x) - f^+, 0)]$ where $f^+ = \max_i y_i$ is the current best observation. Under a GP this has the closed form $\alpha_{\text{EI}}(x) = \sigma(x)[\gamma\Phi(\gamma) + \phi(\gamma)]$ with $\gamma = (\mu(x) - f^+)/\sigma(x)$ (Frazier, 2018). EI is positive even when $\mu(x) < f^+$, provided $\sigma(x)$ is large—this is why EI naturally explores uncertain regions. EI has no finite-time regret guarantee.

Upper Confidence Bound (UCB). $\alpha_{\text{UCB}}(x) = \mu(x) + \sqrt{\beta} \sigma(x)$. The parameter β directly controls the exploration–exploitation trade-off. Under a theory-motivated schedule $\beta_t \propto \log t$, GP-UCB achieves sub-linear cumulative regret (Frazier, 2018). In practice a fixed β is common; I use $\beta = 2$.

Pure-exploration strategies. Maximizing $\sigma(x)$ alone—without any $\mu(x)$ term—corresponds to selecting the point about which the GP is most uncertain. This pure-exploration strategy is useful as a controlled ablation: it uses the GP’s calibration machinery but discards all exploitation signal. Blind random search serves as the weakest baseline. De Ath et al. 2019 give a theoretical framing: they show EI and UCB both lie on the exploration–exploitation Pareto front in the (μ, σ) plane, whereas maximum- σ sampling is off this front—it is Pareto-dominated in the exploitation dimension.

Their empirical results suggest this off-front position is consequential in higher dimensions.

2.4 Related Work

Comparative benchmarks. Several large-scale BO benchmarks have compared acquisition strategies. Shahriari et al. 2016 review EI, UCB, Thompson Sampling, and Entropy Search across diverse tasks, and conclude that the choice of surrogate model typically matters more than the fine-grained choice of acquisition function—a claim my normalization experiment instantiates concretely. De Ath et al. 2019 compare standard acquisition functions against ϵ -greedy methods on ten synthetic benchmarks plus real-world tasks, finding that simple greedy strategies are competitive with EI and UCB in high dimensions.

Statistical methodology. Demšar (2006) argues that comparing classifiers—or, by extension, optimization algorithms—over multiple datasets requires non-parametric rank-based tests rather than mean-performance comparisons, because regret distributions are typically non-Gaussian and the null hypothesis of equal distributions is not testable with a single paired t -test across datasets. I adopt the Friedman–Nemenyi framework he recommends for exactly this reason.

3 Methods

3.1 Bayesian Optimization Protocol

I follow the standard sequential BO protocol. An initial design of $n_{\text{init}} = 2d$ points is drawn via Sobol quasi-random sampling, where d is the problem dimensionality; Sobol provides more uniform space-filling coverage than independent uniform random draws. After the initial design I run $T = 20$ sequential optimization iterations. At each step the GP surrogate is refitted to all accumulated observations, the acquisition function is maximized to select one candidate point, and the true objective is evaluated at that point. All tensor computations use `torch.float64` to preserve GP numerical stability.

Performance metric. Performance is measured by *simple regret*

$$r_t = f^* - \max_{i \leq t} y_i,$$

where f^* is the known optimal value and y_i are observed function values. Because all six problems are framed as *maximization*, $r_t \geq 0$ throughout every run and decreases toward zero as the optimizer improves.

Experimental protocol. Each strategy–problem pair is evaluated across 10 independent random seeds. Acquisition functions are maximized using BoTorch’s `optimize_acqf`

with multi-start optimization (`num_restarts = 10`, `raw_samples = 64`).

3.2 Gaussian Process Surrogate

I use BoTorch’s `SingleTaskGP` (Balandat et al., 2020) as the surrogate: a Gaussian process with a radial basis function (RBF) kernel equipped with automatic relevance determination (ARD), one independent lengthscale per input dimension. Kernel hyperparameters—lengthscales, output scale, and noise variance—are fitted by maximizing the exact marginal log-likelihood (MLL) at every iteration.

Input normalization. A critical design choice is to normalize GP inputs to the unit hypercube $[0, 1]^d$ and standardize outputs to zero mean and unit variance before fitting. I apply BoTorch’s `Normalize(d, bounds=problem.bounds)` and `Standardize(m=1)` transforms, using the *known, fixed* problem bounds rather than data-adaptive bounds. Fixing the normalization range to the full search space prevents it from drifting as observations accumulate, giving the GP a stable length-scale calibration throughout the run.

The necessity of this choice was revealed empirically by the two engineering problems. Borehole and Piston have inputs spanning approximately seven orders of magnitude (e.g. Piston’s initial gas volume $V_0 \in [0.002, 0.010] \text{ m}^3$ alongside ambient pressure $P_0 \in [90\,000, 110\,000] \text{ Pa}$, a factor of roughly 10^7). Without normalization, the GP cannot learn sensible per-dimension lengthscales across these scales, causing EI and UCB to degrade to random-search performance. Table 1 shows the effect on Piston, where the controlled comparison is cleanest: adding normalization reduces EI’s mean final regret from 0.41 to 0.019 ($\approx 20\times$) and UCB’s from 0.39 to 0.0019 ($\approx 200\times$), while the Random baseline—which makes no GP calls—remains unchanged at ≈ 0.40 . The unchanged Random performance isolates the fault squarely to the surrogate, not the problem formulation.

Table 1: Effect of GP input normalization on mean final regret on Piston (7D), 10 seeds. Random is unchanged because it does not use the GP, confirming the fault is in the surrogate.

Strategy	Before norm.	After norm.	Improvement
EI	0.41	0.019	$\approx 20\times$
UCB	0.39	0.0019	$\approx 200\times$
Random	0.40	0.41	—

3.3 Acquisition Strategies

I benchmark four strategies that together span the full exploration–exploitation spectrum (Table 2).

Expected Improvement (EI).

$$\alpha_{\text{EI}}(x) = \mathbb{E}[\max(f(x) - f^+, 0)],$$

where $f^+ = \max_i y_i$ is the current best observed value. Under a GP surrogate this has the closed form $\alpha_{\text{EI}}(x) = \sigma(x)[\gamma(x)\Phi(\gamma(x)) + \phi(\gamma(x))]$, with $\gamma(x) = (\mu(x) - f^+)/\sigma(x)$, where Φ and ϕ are the standard Gaussian CDF and PDF. EI rewards both a high posterior mean (exploitation) and a high posterior variance (exploration); neither component alone is sufficient to generate positive EI. Implemented via `botorch.acquisition.ExpectedImprovement` with `best_f = train_y.max()`.

Upper Confidence Bound (UCB).

$$\alpha_{\text{UCB}}(x) = \mu(x) + \sqrt{\beta} \sigma(x), \quad \beta = 2.$$

The β parameter controls the exploration–exploitation trade-off. Unlike EI, GP-UCB has finite-time regret guarantees under appropriate β schedules (Frazier, 2018). Implemented via `botorch.acquisition.UpperConfidenceBound`.

Uncertainty (pure exploration).

$$\alpha_{\text{Unc}}(x) = \sigma(x).$$

This strategy selects the point of maximum posterior standard deviation, entirely ignoring the posterior mean. It is the pure-exploration extreme of the spectrum: it uses the same GP as EI and UCB but discards all exploitation signal. Its inclusion allows me to isolate whether the GP’s uncertainty estimate *alone*—without any mean information—is sufficient to outperform blind search. Implemented via `botorch.acquisition.PosteriorStandardDeviation`.

Random (baseline). Points are drawn from a Sobol sequence within the search bounds at every iteration; the GP is never called. This is the weakest baseline that any data-efficient strategy must surpass.

Table 2: The four acquisition strategies and their position on the exploration–exploitation spectrum.

Strategy	Acquisition	Uses μ ?	Uses σ ?
EI	ExpectedImprovement	✓	✓
UCB	UpperConfidenceBound	✓	✓
Uncertainty	PosteriorStdDev		✓
Random	Sobol sampling		

3.4 Problem Suite

I benchmark on six functions spanning dimensions $d \in \{2, 6, 7, 8, 10\}$ across two categories (Table 3).

Synthetic problems (Branin, Six-Hump Camel, Hartmann-6, Ackley-10D) wrap BoTorch’s built-in test functions with `negate=True` to convert minimization to maximization; optimal values are analytically known. The Ackley bounds are narrowed from BoTorch’s default $[-32.768, 32.768]^{10}$ to the literature-standard $[-5, 5]^{10}$.

Engineering problems model physical systems with inputs at physically meaningful, heterogeneous scales. *Borehole* (8D) models water flow through a borehole; inputs include borehole radius, hydraulic transmissivities, water-table elevations, and aquifer conductivity, sourced from SMT’s *WaterFlow*. *Piston* (7D) models piston cycle time as a function of mass M , surface area S , initial gas volume V_0 , spring stiffness k , atmospheric pressure P_0 , ambient temperature T_a , and gas temperature T_0 (Surjanovic and Bingham, 2013). Piston is absent from the SMT version used, so it is implemented manually from the standard closed-form formula. For both engineering functions the global maximum lies at a box vertex; optimal values are obtained by exhaustive corner evaluation ($2^8 = 256$ and $2^7 = 128$ vertices respectively) and a small conservative buffer to guarantee non-negative simple regret.

3.5 Statistical Methodology

With four strategies and six problems, I require a comparison framework that is robust to non-Gaussian, heavy-tailed regret distributions and avoids the inflated type-I error of repeated pairwise t -tests. Following Demšar (2006), I use the Friedman rank test with Nemenyi post-hoc pairwise comparisons.

Friedman test. Each (problem, seed) pair forms one block; the four strategies are ranked by final regret within each block, rank 1 indicating best. This yields $N = 60$ blocks (6 problems $\times$ 10 seeds) and $k = 4$ strategies. The Friedman statistic tests the null hypothesis that all strategies have equal expected rank.

Nemenyi post-hoc test. On rejection of the Friedman null, all $\binom{4}{2} = 6$ pairwise comparisons are performed using Nemenyi’s test at $\alpha = 0.05$ (Terpilowski, 2019). Two strategies are deemed *statistically indistinguishable* when their average-rank difference is less than the critical difference

$$\text{CD} = q_\alpha \sqrt{\frac{k(k+1)}{6N}} = q_{0.05} \sqrt{\frac{4 \times 5}{6 \times 60}},$$

where q_α is the critical value from the Studentized range distribution (divided by $\sqrt{2}$); $\text{CD} \approx 0.61$ for $k = 4$, $N = 60$, $\alpha = 0.05$. Results are visualized as a Critical Difference (CD) diagram (Figure 1), where horizontal bars connect strategies that are not significantly different from one another. The CD value is computed analytically inside the plotting function to remain correct under any change to N or k .

Table 3: Benchmark problem suite. All problems are framed as maximization; simple regret is $f^* - \text{best observed}$. Engineering optimal values are derived from exhaustive corner search (2^d vertices) rather than random sampling, which systematically undershoots vertex maxima on monotone functions (see Section 3.2).

Problem	Type	d	Domain	f^*
Branin	Synthetic	2	standard	-0.3979
Six-Hump Camel	Synthetic	2	$[-3, 3] \times [-2, 2]$	1.0316
Hartmann-6	Synthetic	6	$[0, 1]^6$	3.3224
Piston	Engineering	7	multi-scale	1.20
Borehole	Engineering	8	multi-scale	310.0
Ackley-10D	Synthetic	10	$[-5, 5]^{10}$	0.0

4 Results

I report three complementary views of the benchmark: the omnibus statistical ranking, per-problem median regret, and regret trajectories. Together they tell a consistent story: the 240 BO runs partition the four strategies into two statistically separated equivalence classes.

4.1 Omnibus Statistical Test

The Friedman test strongly rejects the null hypothesis of equal expected ranks across the four strategies: $\chi^2(3) = 128.82$, $p = 9.73 \times 10^{-28}$ ($N = 60$ blocks, $k = 4$). Average ranks are UCB = 1.39, EI = 1.76, Uncertainty = 3.38, Random = 3.48.

The rank ordering partitions into two well-separated groups. The gap between EI (1.76) and Uncertainty (3.38) is 1.62, more than two and a half times the critical difference $CD = 0.61$. Within each group the gaps are small: UCB and EI differ by 0.37 (below CD), and Uncertainty and Random differ by 0.10 (far below CD).

4.2 Nemenyi Pairwise Comparisons

All six pairwise Nemenyi comparisons at $\alpha = 0.05$ are reported in Table 4 and visualized in the Critical Difference diagram (Figure 1).

The results define two equivalence classes:

- **Balanced strategies:** {UCB, EI} — statistically indistinguishable from each other ($p = 0.404$), both significantly better than the pure-exploration strategies ($p < 10^{-6}$).
- **Pure-exploration strategies:** {Uncertainty, Random} — statistically indistinguishable from each other ($p = 0.974$), both significantly worse than the balanced strategies ($p < 10^{-6}$).

The most striking result is the Uncertainty–Random tie. Uncertainty selects points by maximizing posterior standard deviation using the same fitted GP as EI and UCB; it is not a blind strategy. Yet its rank (3.38) is statistically indistinguishable from Random (3.48, $p = 0.974$). This isolates the contribution of exploitation: the GP’s uncertainty estimate, used *without* the posterior mean, provides no measurable advantage over blind Sobol sampling.

4.3 Per-Problem Median Regret

Table 5 reports median final regret across 10 seeds for each problem–strategy pair. I report median rather than mean because several seeds produce outlier-high regret on certain problem–strategy combinations; for example, EI on Borehole has a mean final regret of 8.63 (inflated by two outlier seeds) but a median of only 2.41. Median is more representative of typical performance and robust to heavy-tailed distributions.

Several patterns are visible across the table. First, the balanced strategies (EI, UCB) lead on every single problem; the pure-exploration strategies (Uncertainty, Random) never win a row. Second, the magnitude of the advantage varies substantially with problem structure. On Borehole (8D, engineering), UCB’s median regret (0.42) is $263\times$ lower than Random’s (111.6), reflecting the structured, near-monotone response surface that a well-calibrated GP can exploit. On Ackley-10D (synthetic, highly multimodal), EI’s median (3.80) is only $2.2\times$ lower than Random’s (8.27), consistent with the curse of dimensionality degrading GP calibration at this budget. Third, among the balanced strategies, neither EI nor UCB consistently dominates: UCB ranks first on four of six problems, EI on two, and their average ranks (1.39 vs. 1.76) are not significantly different (Nemenyi $p = 0.404$).

4.4 Regret Trajectories

Figures 2 and 3 show mean regret trajectories ($\pm$ one standard deviation) across seeds for the two most contrasting problems.

4.5 Effect of Dimensionality

To quantify how the EI–Random gap changes with dimensionality I use mean final regret, which captures tail behavior that median suppresses. On Branin (2D), EI achieves a mean regret of 0.034 against Random’s 1.58, a $\approx 47\times$ advantage. On Ackley-10D, EI achieves 4.05 against Random’s 8.13, a $\approx 2\times$ advantage—a more than 20-fold reduction in relative gain over eight additional dimensions.

This collapse in advantage is consistent with the theoretical analysis in Shahriari et al. (2016): in high dimensions the

Table 4: Nemenyi post-hoc p -values. NS = not significant at $\alpha = 0.05$; *** = $p < 10^{-6}$. Two distinct equivalence classes emerge: {UCB, EI} and {Uncertainty, Random}.

	UCB	EI	Uncertainty	Random
UCB	—	0.404 (NS)	$< 10^{-6}$ ***	$< 10^{-6}$ ***
EI		—	$< 10^{-6}$ ***	$< 10^{-6}$ ***
Uncertainty			—	0.974 (NS)
Random				—

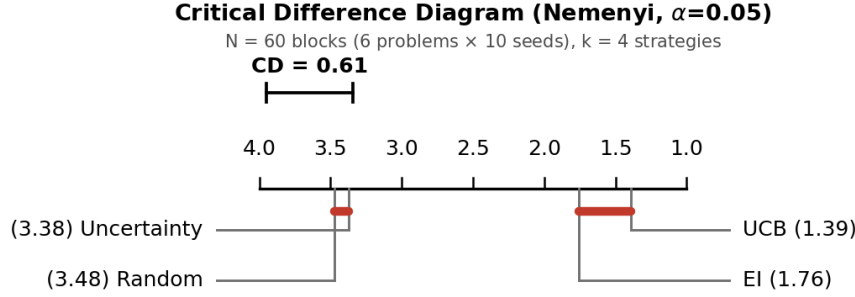


Figure 1: Critical Difference diagram ($\alpha = 0.05$, $k = 4$, $N = 60$, $CD = 0.61$). Horizontal bars connect strategies that are not significantly different. Two separate cliques are visible: {UCB, EI} (left, lower ranks = better) and {Uncertainty, Random} (right, higher ranks = worse).

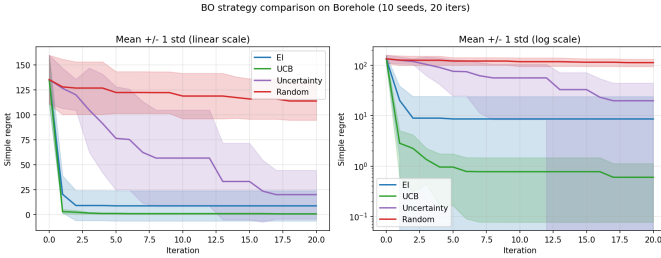


Figure 2: Mean regret $\pm$ std over 20 iterations on Borehole (8D). EI and UCB converge rapidly while Uncertainty and Random show negligible improvement. The large EI std band is driven by two outlier seeds with elevated final regret, inflating the mean well above the median (8.63 vs. 2.41).

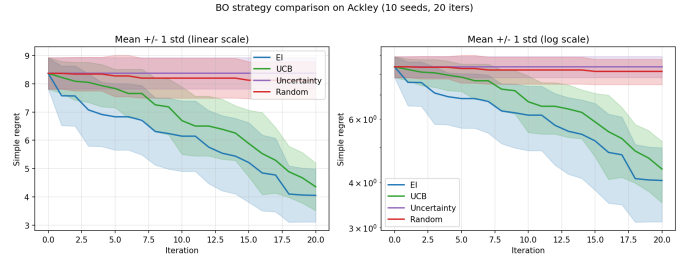


Figure 3: Mean regret $\pm$ std over 20 iterations on Ackley-10D. All four strategies remain at high regret; EI and UCB show a marginal but consistent advantage over pure-exploration strategies. The narrow gap between balanced and pure-exploration strategies illustrates dimensionality-driven degradation of GP calibration.

5 Discussion

5.1 Exploitation Is the Critical Ingredient

GP surrogate requires exponentially more observations to achieve accurate uncertainty estimates, and the 20-iteration budget used here is far from sufficient to cover the $[-5, 5]^{10}$ search space. The Ackley function has no low-effective-dimensionality structure that could be exploited through embeddings or variable selection, making it a worst-case scenario for GP-based BO at this budget.

The most consequential result is not that EI and UCB outperform Random—that is expected—but that Uncertainty is statistically indistinguishable from Random ($p = 0.974$, average ranks 3.38 vs. 3.48). Uncertainty is not a blind strategy: it fits the same GP as EI and UCB at every iteration and selects the point of maximum posterior standard deviation. Yet its rank matches that of blind Sobol

Table 5: Median final simple regret after 20 iterations, 10 seeds. Bold = best per row. Ordered by dimension. All problems are maximization; lower regret is better.

Problem (d)	EI	UCB	Uncertainty	Random
Branin (2)	0.026	0.015	2.676	0.920
SixHumpCamel (2)	0.057	0.287	0.936	0.924
Hartmann-6 (6)	0.183	0.254	1.913	1.802
Piston (7)	0.016	0.001	0.033	0.395
Borehole (8)	2.409	0.424	5.747	111.642
Ackley-10D (10)	3.800	4.295	8.441	8.272
Avg. rank	1.76	1.39	3.38	3.48

sampling across 60 (problem, seed) blocks.

This isolates where the value of BO lies. Uncertainty and Random share one characteristic: neither uses the GP’s posterior mean. EI and UCB share the complementary characteristic: both combine the posterior mean with the posterior variance. The equivalence classes therefore provide a clean experimental decomposition—the GP’s uncertainty estimate alone is insufficient; exploitation of the mean is the necessary component.

Two independent lines of theory predict this outcome. De Ath et al. 2019 recast acquisition function selection as a bi-objective problem: they show that EI and UCB both select points on the exploration–exploitation Pareto front, whereas a strategy that uses only $\sigma(x)$ (the pure-exploration extreme) sits off this front in the exploitation dimension. Shahriari et al. 2016 observe that acquisition functions matter, but caution against over-indexing on any single one; their framing implies that the *balance* between the two GP signals is what separates effective from ineffective acquisition strategies. Both predictions are borne out by the Uncertainty–Random equivalence.

5.2 EI and UCB Are Interchangeable at This Budget

Within the balanced-strategy class, EI and UCB are statistically indistinguishable (Nemenyi $p = 0.404$), despite different theoretical properties: UCB has finite-time regret guarantees under appropriate β schedules, whereas EI has no such guarantee. The per-problem medians echo this: UCB leads on four of six problems, EI on two, with no consistent pattern by problem type or dimension.

De Ath et al. 2019 provide a mechanistic explanation: both EI and UCB lie on the exploration–exploitation Pareto front, so they explore at similar rates; any residual difference is smaller than the problem-to-problem and seed-to-seed variance at a 20-iteration budget. Shahriari et al. 2016 reinforce this with a broader observation: the surrogate model’s adequacy typically dominates the fine-grained acquisition choice. The EI–UCB equivalence is therefore not a sample-size failure but the expected outcome of using two well-designed balanced strategies on the same adequate surrogate.

5.3 GP Normalization as a First-Class Methodological Result

The GP normalization result warrants discussion beyond a footnote in the Methods section. The initial experiment on Piston (7D) showed EI and UCB performing at random-search level—plausible only if the surrogate were failing. Piston’s inputs span approximately seven orders of magnitude—initial gas volume $V_0 \in [0.002, 0.010] \text{ m}^3$ alongside ambient pressure $P_0 \in [90\,000, 110\,000] \text{ Pa}$ —so an un-normalized GP cannot learn meaningful per-dimension lengthscales. Adding **Normalize** and **Standardize** recovered $\approx 20\times$ improvement for EI and $\approx 200\times$ for UCB, while Random—which bypasses the GP—remained unchanged, isolating the fault to the surrogate.

This finding directly instantiates Shahriari et al.’s 2016 claim that surrogate adequacy dominates acquisition choice: the problem was not that EI or UCB was the wrong acquisition function, but that the surrogate itself was miscalibrated. It also underscores a practical lesson for engineers deploying BO on real systems: physical inputs with heterogeneous scales require explicit normalization to the known search bounds, not data-adaptive bounds that drift as observations accumulate.

5.4 Limitations

Budget. The 20-iteration budget is appropriate for expensive black-box functions (Frazier, 2018), but it is tight relative to the 10D Ackley search space. At this budget, no strategy explores more than a small fraction of $[-5, 5]^{10}$, which suppresses the EI–Random advantage from $47\times$ on 2D Branin to $2\times$ on 10D Ackley and limits conclusions about asymptotic performance. Longer-horizon experiments would be needed to determine whether balanced strategies accumulate a growing advantage in high dimensions or plateau.

Fixed β for UCB. I use a fixed $\beta = 2$ throughout, which does not follow the theory-motivated schedule $\beta_t \propto \log t$ under which GP-UCB’s regret guarantees hold (Frazier, 2018). A theory-consistent β schedule may widen or narrow

the EI-UCB gap; I cannot rule out this effect from the current experiment.

Sequential-only evaluation. All results use sequential ($q = 1$) BO. Batch acquisition ($q > 1$, e.g. qEI) is important in practice when multiple evaluations can run in parallel, and the relative performance of strategies may differ in the batch setting.

Synthetic and physics-emulator problems only. The benchmark includes six analytic or closed-form problems. Real engineering simulations—such as CFD or aerodynamic design codes—may exhibit noise, discontinuities, or constraint violations not present here. Whether the two-clique finding persists on black-box simulation outputs remains an open question.

6 Conclusion

I benchmarked four acquisition strategies—EI, UCB, Uncertainty, and Random—on six problems spanning dimensions 2 to 10 and two problem families (synthetic and engineering), using a Friedman–Nemenyi non-parametric framework with $N = 60$ blocks. The results converge on two findings.

First, the 240 BO runs partition cleanly into two equivalence classes: balanced strategies {EI, UCB} are statistically superior to pure-exploration strategies {Uncertainty, Random}, with all cross-class Nemenyi p -values below 10^{-6} and an inter-class rank gap of $1.62 \gg \text{CD} = 0.61$. The Uncertainty–Random tie ($p = 0.974$) is the sharpest statement of this result: using a GP for exploration only, without any exploitation of the posterior mean, provides no measurable advantage over blind random search.

Second, GP normalization to the known problem bounds is a first-class engineering requirement on multi-scale problems. Without it, BO on Piston degraded to random-search performance; restoring it recovered one to two orders of magnitude in final regret. This confirms the theoretical expectation that surrogate adequacy dominates the acquisition-function choice and highlights a pitfall specific to physical systems with heterogeneous input scales.

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